Project Abstract :

HealTrack is a **portable IoT-based health monitoring system** built using **ESP32, MAX30102, DS18B20, and an OLED display**. It continuously measures **heart rate, oxygen saturation (SpO_2), and body temperature.**

Problems it solves :

HealTrack solves this by offering a **low-cost, portable, and real-time solution** that anyone can use at home

Objective :

To provide basic health insights/advice based on threshold values (e.g., high HR, low SpO_2, fever).

Innovation in HealTrack :

All-in-one portable device – Combines heart rate, oxygen saturation, and temperature in a single low-cost system.

Uniqeness :

Unlike simple pulse oximeter modules, **HealTrack combines multiple vitals + advice + IoT** connectivity and It offers a **portable wellness solution** that can be used at home, in education, or as a prototype for wearable devices.

Sustainability & Eco-Friendliness :

Low-power design with rechargeable battery,
- Modular & reusable components,
- Supports solar/USB charging,
- Digital data → paperless tracking,
- Compact build with eco-friendly casing options.

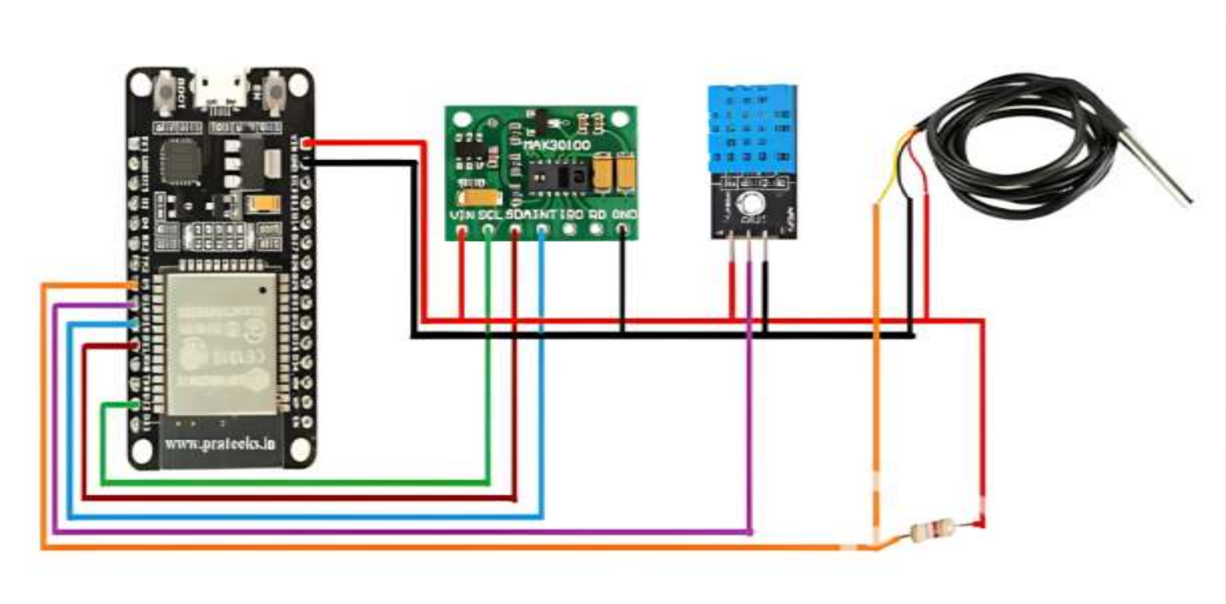
Different Parameters Measured by HealTrack :

Parameter	Source	How it's obtained
Heart Rate (bpm)	MAX30100	Peak detection on PPG waveform
SpO_2 (%)	MAX30100	Ratio-of-ratios using IR/Red
Pulse Waveform (BVP)	MAX30100	Raw/filtered PPG signal
Perfusion Index (PI)	MAX30100	AC/DC of PPG
Respiratory Rate (rpm)	MAX30100 (derived)	Respiratory-induced modulation of PPG (RIAV/RIIV/RIIVD)
digital temperature and humidity	DHT11	a single-wire digital protocol
Skin / Ambient Temperature (°C)	DS18B20	1-wire readout

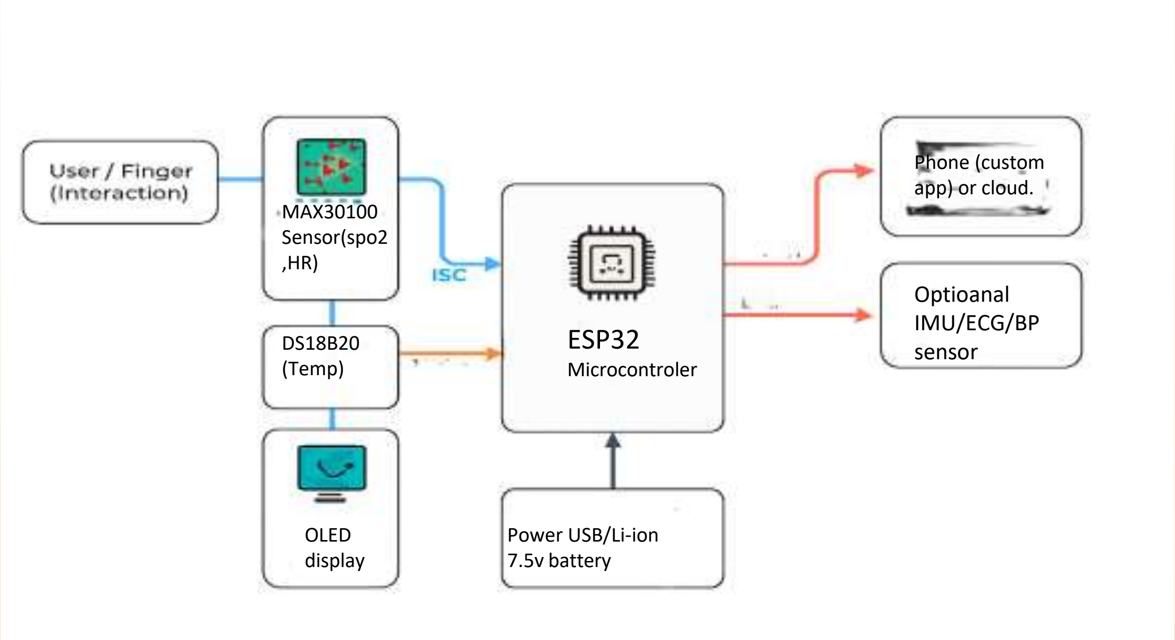
Circuit Componants and Billing :

S.N	Components	Quantity	Cost(rupee)
1	ESP32 Board	1	
2	MAX30100 Sensor	1	
3	DHT11 Sensor	1	
4	DS18B20	1	
5	5v Power Supply	1	
6	Zero PCB	1	

Circuit Diagram :



System Block Diagram :



Working Model (Prototype Description) :

1. Mount ESP32 on a small breadboard or prototyping PCB.
2. Place MAX30100 breakout so a finger can rest on its sensor window.
3. Fix the DS18B20 in contact with skin (or expose to ambient if measuring environment).
4. Place OLED where the user can view the readings easily.
5. Power from USB for bench testing; use Li-ion + TP4056 for mobile demo.

User interaction:

The user places their finger on the MAX30100 for ~10–20 sec. The device collects and displays HR, SpO_2, and temperature, while also sending data to the paired phone app. Use the OLED to show a simple dashboard and an SQI bar so the user knows if the reading is reliable.

Monetization & Business Potential :

Target Users :

Students & researchers (affordable DIY health device),
Fitness enthusiasts (portable vitals tracker),
Rural health workers (low-cost alternative to bulky medical devices).

Revenue Models :

Device Sales: Affordable hardware kit ,
Subscription App: Premium features like data history, trends, cloud backup,
Licensing: Offer design as a low-cost solution for NGOs / telemedicine startups,
Customization: Branded devices for fitness centers, schools, or clinics.

Market Potential :

Growing demand for **low-cost wearable health monitors** in India & globally
IoT healthcare market projected to grow at 15–20% CAGR
Huge opportunity in **preventive health monitoring** (post-COVID awareness)

Project Status :

Completed Prototype

Future Improvements & Expansion :

1. Blood Pressure (NIBP/PPG-based) → expand into cardiovascular monitoring
2. **AI/ML** models for predicting health risks (stress, fatigue, irregular heart patterns)